

Question number	Answer	Additional guidance	Mark
5(a)	An explanation that includes three of the following points: • active site no longer complementary to trans-retinal (1) • cis-retinal cannot be regenerated (1) • therefore, rhodopsin cannot be formed (1) • and rod cells will not have any photoreceptor (1)		(3)

Question number	Answer	Additional guidance	Mark
5(b)	An answer that includes the following points: • increasing the dose of gene therapy increased cis-retinal in the eye (1) • the RPE65 gene enables the conversion of trans-retinal to cis-retinal (1) • increasing the dose of gene therapy increased the ability to follow a path (1) • the production of cis-retinal restores vision (1) • reference to error bars used to support conclusions (1)	Accept increasing the dose of gene therapy does not alter trans-retinal very much in the eye	(4)

Question number	Answer	Additional guidance	Mark
5(c)	A description that includes the following points: • use microarray techniques to identify genes that are actively transcribed in the eye (1) • sequencing the genome of individuals with inherited forms of blindness (1) • compare the sequences for active genes from a large number of individuals with and without blindness to identify mutations (1)		(3)

